

EXPERIMENT 10

Aim: Explore the Cryptocurrency Landscape

Theory:

Cryptocurrency Landscape –

Theory :

Cryptocurrencies are decentralized digital assets that operate on blockchain technology, enabling secure, transparent, and immutable transactions without the need for a central authority. These systems rely on distributed ledgers where transactions are validated through consensus mechanisms and recorded across a network of nodes.

Data Sources

Reliable and diverse data sources are essential for understanding cryptocurrency markets:

- Financial Data Platforms – Services like CoinGecko and CoinMarketCap provide structured data such as market capitalization, circulating supply, trading volume, and historical price trends.
- News Aggregators – Platforms such as CryptoPanic collect real-time news from multiple sources, helping track major events impacting the market.
- Social Sentiment Analytics – Tools like LunarCrush analyze social media activity to measure public sentiment and engagement.
- Regulatory Sources – Official portals like the U.S. Securities and Exchange Commission and frameworks such as MiCA provide insights into legal policies and compliance requirements.

Types of Cryptocurrencies

The cryptocurrency ecosystem consists of various categories based on functionality:

- Store of Value (SoV) – Bitcoin (BTC), known for limited supply and long-term value preservation.
- Utility Tokens – Ether (ETH) and Chainlink (LINK), used for network operations and smart contract execution.
- Stablecoins – Tether (USDT) and USD Coin (USDC), designed to maintain price stability by being pegged to fiat currencies.
- Governance Tokens – Uniswap (UNI) and Aave (AAVE), enabling users to participate in decentralized decision-making.

Key Characteristics

Cryptocurrencies exhibit the following features:

- Decentralization – No central authority controls the system.
- Transparency – Transactions are publicly recorded on the blockchain.
- Immutability – Data cannot be altered once confirmed.
- Security – Cryptographic algorithms ensure data integrity and safety.

Market Dynamics

The cryptocurrency market is influenced by multiple factors:

- Volatility – Prices fluctuate significantly due to demand, sentiment, and external events.
- Liquidity – Depends on trading volume and market participation.
- Market Sentiment – News, social media, and investor behavior impact price trends.
- Regulation – Government policies and frameworks influence adoption and stability.

Blockchain Technology Advancements

Modern blockchain systems are evolving to overcome limitations:

- Consensus Mechanisms – Shift from Proof of Work (PoW) to Proof of Stake (PoS) for efficiency.
- Scalability Solutions – Layer-2 technologies and sharding improve transaction speed.
- Interoperability – Enables communication between different blockchain networks.
- Privacy Enhancements – Zero-Knowledge Proofs (ZKP) allow secure verification without revealing sensitive data.

Adoption and Emerging Trends

Cryptocurrency adoption continues to grow across sectors:

- Digital Payments – Companies like Stripe and Shopify support crypto transactions.
- Decentralized Finance (DeFi) – Financial services operate without intermediaries.
- Non-Custodial Wallets – MetaMask and Phantom give users full asset control.
- Asset Tokenization – Real-world assets are increasingly represented on blockchain networks.

Utilizing Api for analysis and visualization :

Code:

```
!pip install requests pandas matplotlib

# Imports
import requests
import pandas as pd
import matplotlib.pyplot as plt

# ----- CONFIG -----
API_KEY = "api-key"
COIN = "BTC"
CURRENCY = "USD"
DAYS = 180

# ----- FETCH FUNCTION -----
def fetch_data(coin, currency, days):
    url = "https://min-api.cryptocompare.com/data/v2/histoday"

    params = {
        "fsym": coin,
        "tsym": currency,
        "limit": days,
        "api_key": API_KEY
    }

    res = requests.get(url, params=params)
    data = res.json()

    if data["Response"] != "Success":
        print("Error:", data.get("Message"))
        return None

    return data["Data"]["Data"]

# ----- LOAD DATA -----
raw = fetch_data(COIN, CURRENCY, DAYS)
df = pd.DataFrame(raw)
```

```
# Time formatting
df["date"] = pd.to_datetime(df["time"], unit="s")
df.set_index("date", inplace=True)

# Core columns
df["price"] = df["close"]
df["volume"] = df["volumeto"]

# ----- CALCULATIONS -----
df["returns"] = df["price"].pct_change()
df["MA_7"] = df["price"].rolling(7).mean()
df["MA_21"] = df["price"].rolling(21).mean()
df["cumulative_return"] = (1 + df["returns"]).cumprod()
df["volatility"] = df["returns"].rolling(7).std() * 100

# ----- PLOTTING (6 GRAPHS) -----
plt.figure(figsize=(16, 12))

# 1. Price Trend
plt.subplot(3, 2, 1)
plt.plot(df["price"])
plt.title("1. Price Trend")
plt.grid()

# 2. Price vs Volume
plt.subplot(3, 2, 2)
plt.plot(df["price"], label="Price")
plt.bar(df.index, df["volume"], alpha=0.3, label="Volume")
plt.legend()
plt.title("2. Price vs Volume")

# 3. Returns Distribution
plt.subplot(3, 2, 3)
df["returns"].dropna().hist(bins=40)
plt.title("3. Returns Distribution")

# 4. Moving Averages
plt.subplot(3, 2, 4)
plt.plot(df["price"], label="Price", alpha=0.5)
```

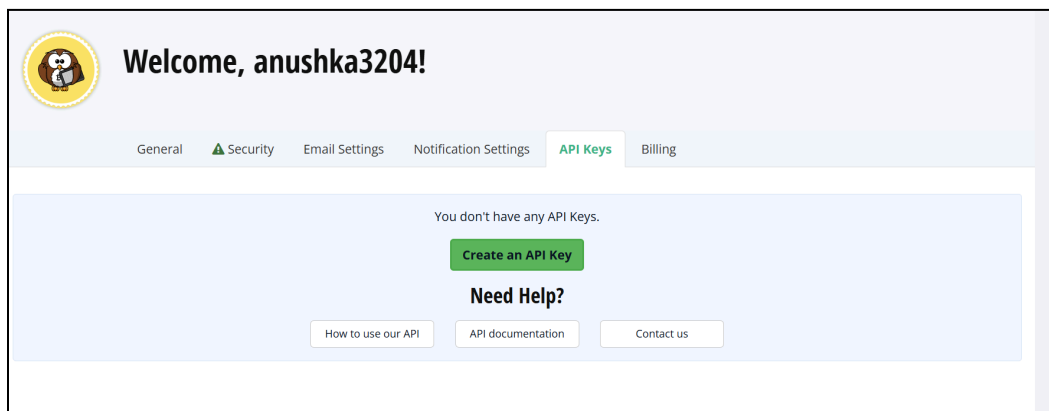
```
plt.plot(df["MA_7"], label="7-Day MA")  
plt.plot(df["MA_21"], label="21-Day MA")  
plt.legend()  
plt.title("4. Moving Averages")
```

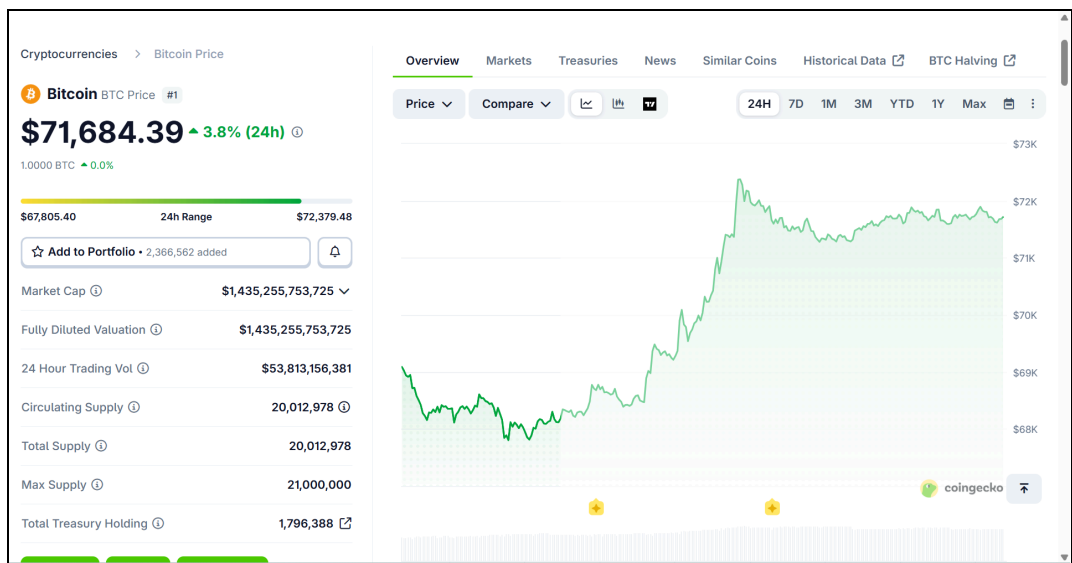
```
# 5. Cumulative Growth  
plt.subplot(3, 2, 5)  
plt.plot(df["cumulative_return"])  
plt.title("5. Cumulative Returns")
```

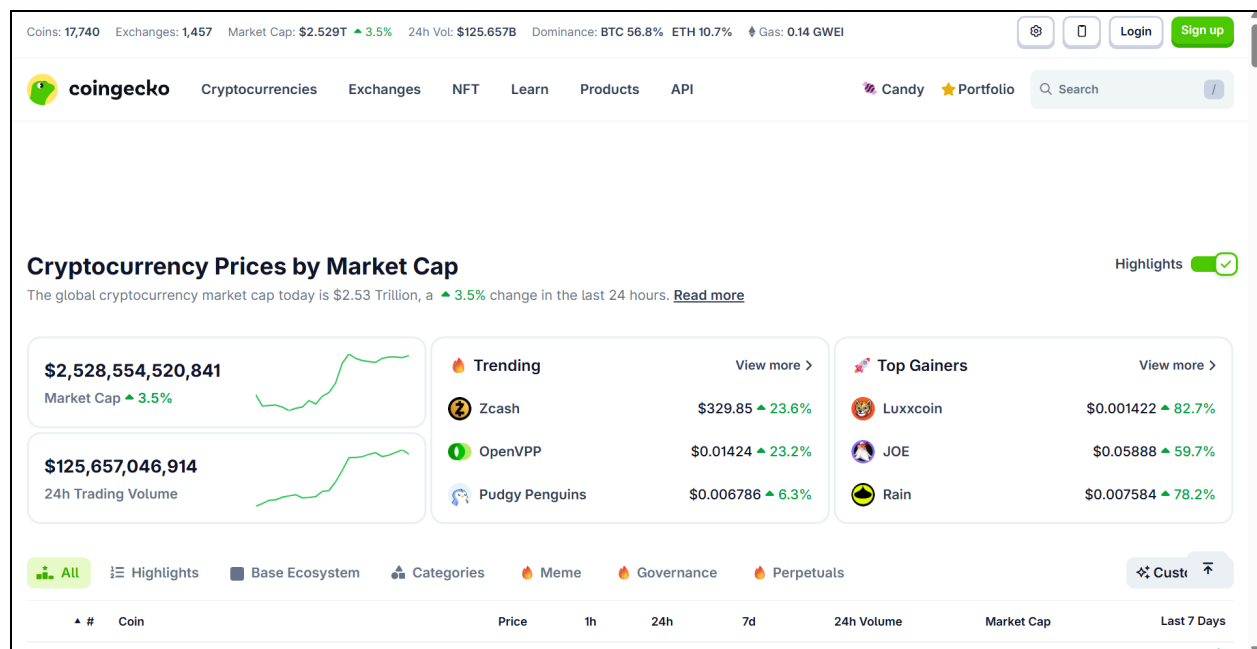
```
# 6. Volatility  
plt.subplot(3, 2, 6)  
plt.plot(df["volatility"])  
plt.title("6. Volatility (%)")
```

```
plt.tight_layout()  
plt.show()
```

Output:-
Coingecko website







CoinMarketCap website

All Networks

BSC

Solana

Base

Ethereum

More

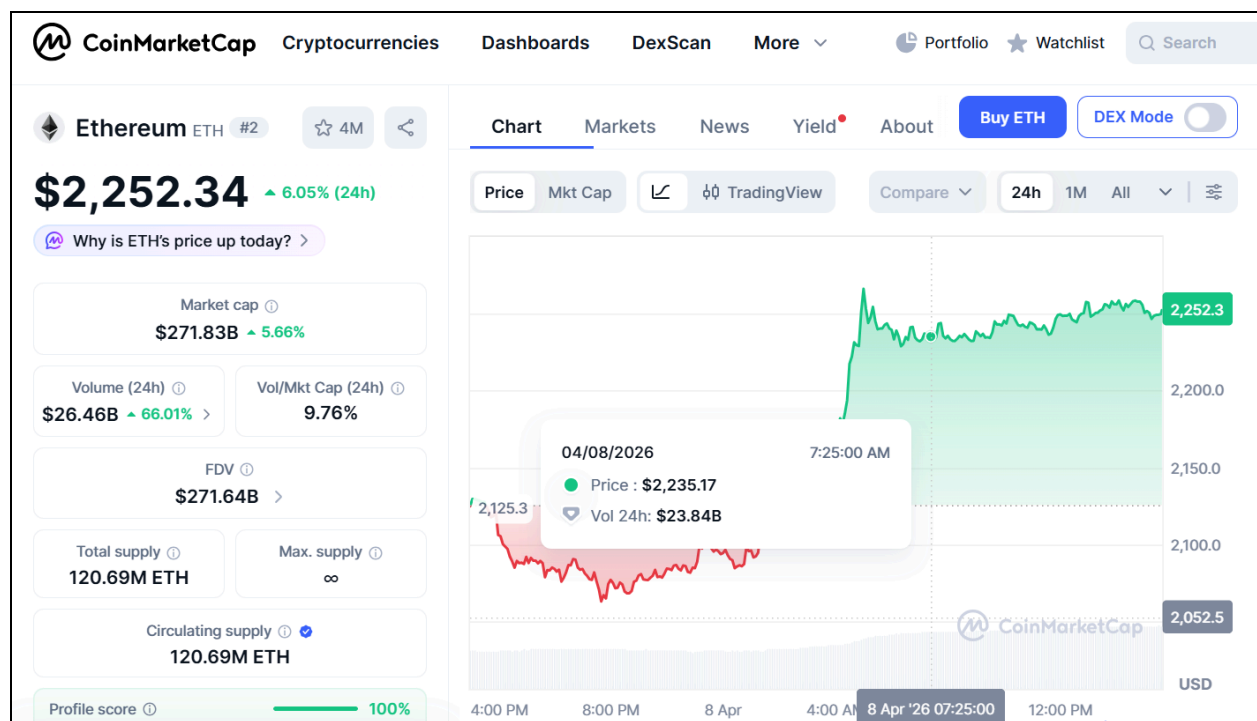
Market Cap

Volume(24h)

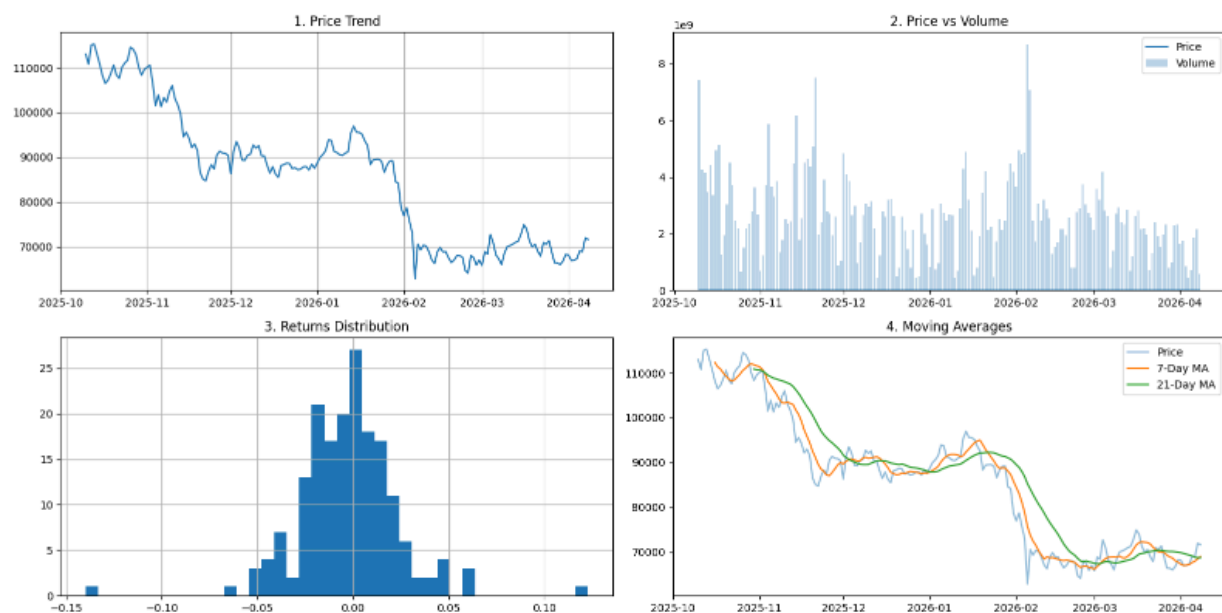
Filters

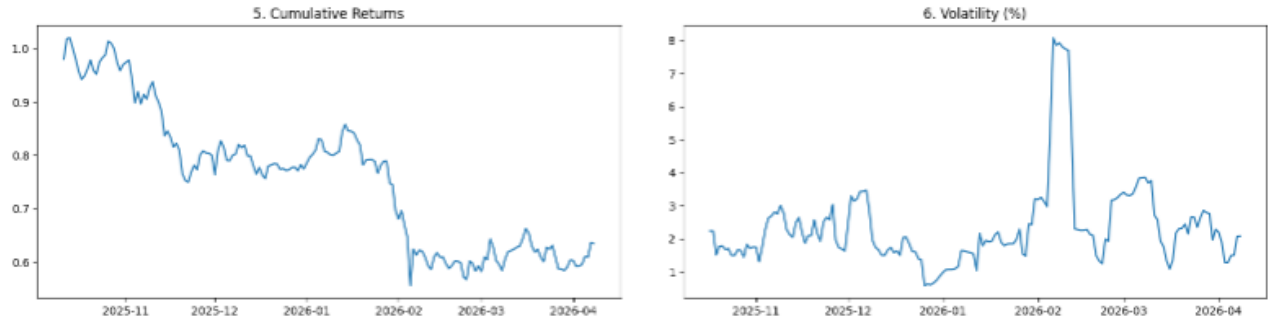
Columns

#	Name		Price	1h	24h	7d	Market Cap	Volume(24h)	Circulating Supply
☆ 🔊	CoinMarketCap 20 Index DTF	CMC20 Buy	\$147.34	▲0.15%	▲4.37%	▲4.13%	\$15,202,093	\$7,247,000 49.31K	103.17K CMC20
☆ 1	Bitcoin BTC	Buy	\$71,703.93	▼0.32%	▲3.88%	▲4.65%	\$1,435,009,321,788	\$50,465,798,161 704.11K	20.01M BTC
☆ 2	Ethereum ETH	Buy	\$2,251.70	▼0.44%	▲6.04%	▲5.63%	\$271,760,366,637	\$26,468,410,099 11.76M	120.69M ETH
☆ 3	Tether USDT	Buy	\$0.9998	▼0.00%	▼0.01%	▲0.02%	\$184,127,697,752	\$99,772,434,008 99.78B	184.15B USDT
☆ 4	XRP XRP	Buy	\$1.38	▼0.05%	▲4.73%	▲2.31%	\$84,860,182,100	\$2,939,951,414 2.12B	61.4B XRP
☆ 5	BNB BNB	Buy	\$612.04	▼0.21%	▲1.24%	▼0.48%	\$83,456,440,261	\$2,210,011,624 3.61M	136.35M BNB
☆ 6	USDC USDC	Buy	\$0.9998	▲0.00%	▼0.01%	▼0.00%	\$77,898,265,470	\$13,428,745,336 13.43B	77.92B USDC
☆ 7	Solana SOL	Buy	\$84.67	▼0.19%	▲5.55%	▲1.63%	\$48,579,699,192	\$6,072,500,218 71.71M	573.7M SOL
☆ 8	TRON TRX	Buy	\$0.3163	▲0.02%	▲0.42%	▲0.44%	\$29,983,968,763	\$531,565,296 1.68B	94.76B TRX
☆ 9	Dogecoin DOGE	Buy	\$0.09506	▲0.22%	▲4.20%	▲2.95%	\$16,109,603,326	\$1,751,129,930 18.48B	153.74B DOGE
☆ 10	Hyperliquid HYPE	Buy						\$346,165,100 8.77M	256.09M HYPE



Dashboard :





Conclusion:-

By integrating automated market data processing, visual trend analysis, blockchain technology evaluation, and adoption metrics, a comprehensive view of the cryptocurrency ecosystem can be achieved. This approach highlights the relationship between blockchain's underlying technical foundations and real-world market behavior, offering valuable insights into current trends as well as potential future developments in the digital asset industry.